ANDY AU

GitHub ◇ LinkedIn ◇ andyau2718@gmail.com ◇ +1 (617) 256 1336

Research Interests

Stochastic optimization and dynamics of learning algorithms; reinforcement learning and stochastic control; Bayesian filtering, robustness, and mathematical foundations of machine learning.

Education

Columbia University Fu Foundation School of Engineering — **New York, NY**
M.S. Artificial Intelligence, Finance & Operations concentration — 09/2026 – 12/2027 (expected)

- **Current graduate study:** Machine Learning for OR & FE; PhD Seminar in Diffusion & Reinforcement Learning; Natural Language Processing; AI for OR & FE.

Boston University College of Arts and Sciences — **Boston, MA**
B.A. Applied Mathematics, minor in Physics | GPA: 3.6/4.0 — 09/2023 – 05/2026

- **Selected coursework:** Stochastic Analysis, Probability Theory, Measure Theory, Stochastic Processes, Time Series & Forecasting, Mathematics of Financial Derivatives, Quantitative Finance.

Preprints & Manuscripts

[1] A. Au. *Path Space Robust Bayesian Portfolio Selection*. *arXiv:2606.24212*, 2026. *Preprint.*

[2] A. Au. *Entropy Regularization as Robustness under Bayesian Drift Uncertainty*. *arXiv:2602.16862*, 2026. *Preprint.*

[3] A. Au. *Temporal Orientation Bias in Adaptive Optimization*. 2026. *Unpublished manuscript.*

Research Experience

Independent Research — Temporal Dependence in Adaptive Optimization — **New York, NY**
Stochastic optimization; RMSprop and Adam under Markovian data — 07/2026 – 10/2026

- Studied how temporal ordering changes the stationary mean offset of adaptive optimizers, comparing dependent data with independent sampling from the same marginal distribution.
- Derived leading-order bias expressions for smooth RMSprop and Adam in scalar quadratic models; identified memory regimes in which an order-dependent offset persists as the learning rate decreases.
- Tested theoretical predictions using 8,192 paired Monte Carlo trajectories per main setting, with zero-effect controls and sensitivity analyses.

Independent Research — Robust Bayesian Portfolio Control — **Boston, MA / New York, NY**
Stochastic control, Bayesian filtering, and relative-entropy ambiguity — 09/2025 – 09/2026

- Modeled adversarial distortions of observed price paths that simultaneously alter trading returns and the Kalman–Bucy drift filter through a shared innovation process.
- Reduced the robust precommitment mean–variance problem to an explicit terminal payoff and replicating policy characterized by a scalar budget equation; established worst-case optimality under a KL penalty.
- For Gaussian drift beliefs, showed that robustness reduces position magnitudes without changing their direction; characterized prior-dependent failures of the nonrobust benchmark.

Research Experience (continued)

Undergraduate Thesis Research **Boston University**
Advisor: Prof. Mark Kon | Bayesian stochastic control 05/2025 – 05/2026

- Derived entropy-regularized, belief-dependent HJB equations for mean–variance portfolio control with an unknown drift learned through continuous-time Bayesian filtering.
- Established Gaussian optimal policies with deterministic Bayesian mean control; showed policy randomization modifies variance rather than information acquisition.

Undergraduate Researcher — Stochastic Machine Learning Group **Boston University**
Advisor: Prof. Julio Castrillón-Candás | KLE residual preprocessing 01/2026 – 05/2026

- Analyzed Karhunen–Loève residual preprocessing for learning signals concentrated in low-variance directions of Gaussian single-index models.
- Used anisotropic SGD theory to characterize effective-dimension and sample-complexity improvements; distinguished these effects from changes to the information exponent.

Research Intern — Deep Learning for Time-Series Forecasting **HKUST, Hong Kong**
Advisors: Prof. Yang Can and Dr. Zhang Fan 07/2024 – 09/2024

- Implemented PatchTST and Temporal Fusion Transformer architectures for sequential financial data and evaluated volatility forecasts against Kalman filter and GARCH baselines.

Selected Computational Research

Reinforcement Learning for Optimal Execution
PPO control; Almgren–Chriss benchmark 05/2026

- Benchmarked reinforcement-learning execution policies against analytical Almgren–Chriss schedules under constant and time-varying volatility.

Risk-Sensitive Deep Hedging
Stochastic control; transaction costs and jump risk 07/2026

- Trained hedging policies with entropic and tail-risk objectives, evaluating hedging error and CVaR under diffusion and jump scenarios.

Transformer-Based Volatility Forecasting
PatchTST ensemble; out-of-sample evaluation 08/2024 – 06/2025

- Developed a 30-model PatchTST ensemble for Bitcoin forward realized-volatility prediction, compared with conventional econometric baselines on held-out chronological data.

Technical Skills

- **Mathematics:** Stochastic calculus and control, HJB equations, Bayesian filtering, stochastic optimization, Markov processes, Monte Carlo methods.
- **Machine learning:** PyTorch, transformer models, reinforcement learning (PPO), regression, XGBoost, time-series forecasting.
- **Programming:** Python (NumPy, pandas, scikit-learn), Git, Linux/Bash, LaTeX.
- **Languages:** English and Mandarin (native); Cantonese (professional).

Honors & Additional Activities

Dean's List, Boston University College of Arts & Sciences. Competitive chess (1700 FIDE); League of Legends (Challenger); Hearthstone (Legend).